

Lecture 17: Nonparametric learning and post-Bayes

Last lecture introduced the **Dirichlet process (DP)**. We begin by recapping some key properties of DP mixture models. We will then put the DP to a new use: using it as a prior over the **population distribution**. We will then sketch how this idea is part of a broader recent trend toward "post-Bayesian" or "predictive Bayesian" approaches.

Recapping DP mixture models

Dirichlet process (DP). A draw $G \sim \text{DP}(\alpha, H)$ is a *discrete* random measure,

$$G = \sum_{k=1}^{\infty} \pi_k \delta_{\phi_k^*}, \quad \phi_k^* \stackrel{\text{iid}}{\sim} H,$$

- The atom locations are random and drawn from the **base measure** H .
- The weights π_k are also random; their spread is controlled by **concentration** α .
- The defining property: for every finite partition $(A_1, \dots, A_K)$ of the space:

$$(G(A_1), \dots, G(A_K)) \sim \text{Dirichlet}(\alpha H(A_1), \dots, \alpha H(A_K)).$$

- Constructively, **stick-breaking** draws the weights:

$$V_k \stackrel{\text{iid}}{\sim} \text{Beta}(1, \alpha), \quad \pi_k = V_k \prod_{j < k} (1 - V_j)$$

The DP mixture model.

$$G \sim \text{DP}(\alpha, H), \quad \phi_i \mid G \stackrel{\text{iid}}{\sim} G, \quad x_i \mid \phi_i \sim f_\phi$$

- Since G is discrete there are **ties** $p(\phi_i = \phi_j) > 0$
- Those ties define the cluster assignments z_i
- Marginalizing out G then defines the **Chinese restaurant process (CRP)**

Conjugacy. With $\phi_{1:n} \mid G \stackrel{\text{iid}}{\sim} G$, the posterior is again a DP:

$$G \mid \phi_{1:n} \sim \text{DP}\left(\alpha + n, \frac{\alpha}{\alpha + n} H + \frac{1}{\alpha + n} \sum_{i=1}^n \delta_{\phi_i}\right).$$

- The posterior **base measure** is the posterior **mean**
- It is a convex combination of the prior base H and the empirical measure $\hat{F}_n$:

$$\mathbb{E}[G \mid \phi_{1:n}] = \frac{\alpha}{\alpha+n} H + \frac{n}{\alpha+n} \hat{F}_n, \quad \hat{F}_n \equiv \frac{1}{n} \sum_{i=1}^n \delta_{\phi_i}.$$

- The **predictive** (integrate G out) is the **Pólya urn** — the CRP stated over values:

$$\phi_{n+1} \mid \phi_{1:n} \sim \frac{\alpha}{\alpha+n} H + \frac{1}{\alpha+n} \sum_{i=1}^n \delta_{\phi_i}.$$

Rethinking Bayesian uncertainty

- Standard Bayes first posits $x \sim f_\phi$, then places a prior on ϕ .
- The uncertainty $p(\phi \mid x_{1:n})$ stems from our uncertainty about ϕ .
- Many appealing properties of standard Bayesian procedures assume that the **true population distribution** F_0 is in the model class---i.e., that the **model is true**:

$$F_0 \in \{f_\phi : \phi \in \Phi\}$$

- This is called the **M-closed** worldview.
- But "*all models are wrong*". In the **M-open** scenario:

$$F_0 \notin \{f_\phi : \phi \in \Phi\}$$

- That said we always have samples of the true **population distribution** $x_i \stackrel{\text{iid}}{\sim} F_0$.
- **Nonparametric learning (NPL)**: Be Bayesian about F_0 : assume the population distribution is is a random measure, and do posterior inference over it.

A DP prior on the population

As is standard in Bayesian nonparametrics (BNP), we used the DP before as a prior over the parameters of a model. Now we will use it to place a prior directly on the **population distribution** of the data, but centered on a parametric model f_ϕ . We will still use the prior $\pi(\phi)$ in a two-level prior called a **mixture of Dirichlet processes** (MDP):

$$\phi \sim \pi(\phi), \quad F \mid \phi \sim \text{DP}(c, f_\phi), \quad x_i \mid F \stackrel{\text{iid}}{\sim} F.$$

- The base measure f_ϕ is **the model you would have used** in standard Bayes

- $\pi(\phi)$ is its usual prior, but also the **mixing distribution** of the MDP
- The concentration c is **how strongly you trust the parametric model**.

By DP conjugacy at each fixed ϕ , and updating ϕ as usual, the posterior is again an MDP:

$$F \mid \phi, x_{1:n} \sim \text{DP}\left(c + n, \frac{c}{c+n} f_\phi + \frac{1}{c+n} \sum_{i=1}^n \delta_{x_i}\right), \quad \phi \mid x_{1:n} \sim \pi(\phi \mid x_{1:n}).$$

- The mixing distribution $\pi(\phi \mid x_{1:n})$ is **exactly the ordinary parametric Bayes posterior**.
- To sample F :
 - **(1)** Draw $\phi \sim \pi(\phi \mid x_{1:n})$
 - **(2)** Draw $F \sim \text{DP}\left(c + n, \frac{c}{c+n} f_\phi + \frac{1}{c+n} \sum_i \delta_{x_i}\right)$.
- We think of this model as **regularizing** the empirical measure $\sum_i \delta_{x_i}$
- More data ($n \gg c$) pulls F toward the data, more trust ($c \gg n$) toward the model.
- We can also think of this as "**un-regularizing**" the parametric posterior to the empirical distribution.

Parameters as functionals of F

- A posterior over the whole distribution F gives a posterior over **any feature** of F .
- Write the parameter of interest as a **functional** — the maximizer of the expected log-likelihood under F :

$$\phi(F) = \arg \max_{\phi} \int \log f_{\phi}(x) dF(x).$$

- Maximizing this expected is the same as minimizing the KL from F to the model:

$$\phi(F) = \arg \min_{\phi} \text{KL}(F \parallel f_{\phi}),$$

- **At the truth** F_0 this is the "pseudo-true parameter" $\phi_0 \equiv \phi(F_0)$.
- If the model is **_correct** (M-closed), then $F_0 = f_{\phi^*}$ and $\phi_0 = \phi^*$ (the KL is zero).
- If the model is **misspecified** (M-open) there is no "true" ϕ and ϕ_0 is the *pseudo-true* parameter (the best KL projection).
- **At the empirical measure** $\hat{F}_n = \frac{1}{n} \sum_i \delta_{x_i}$ this is the MLE:

$$\phi(\hat{F}_n) = \arg \max_{\phi} \int \log f_{\phi} d\hat{F}_n = \arg \max_{\phi} \frac{1}{n} \sum_{i=1}^n \log f_{\phi}(x_i) = \hat{\phi}_{\text{MLE}}.$$

- More generally, replace $\log f_\phi$ by any objective: $\phi(F) = \arg \min_\phi \int \ell(\phi, x) dF(x)$ – a mean, a quantile, a regression coefficient, a decision cost.

Push-forward posterior

- Now in **NPL** we put a **posterior** over F .
- This pushes forward to define a posterior over any functional:

$$F^{(b)} \sim (\text{posterior over } F) \implies \phi^{(b)} = \phi(F^{(b)}).$$

- The samples $\{\phi^{(b)}\}$ are thus a kind of **posterior over ϕ** .
- However the uncertainty no longer stems from ϕ being a random variable in the model.
- Uncertainty about ϕ instead stems (strictly) from our uncertainty about F .
- The model $f_\phi, \pi(\phi)$ still plays a role: it centers the DP prior over F

Computation: the posterior bootstrap

- How do we actually draw F from the DP posterior and evaluate $\phi(F)$?
- **A DP on finitely many atoms is just a Dirichlet over their weights.**
- If the (unnormalized) base measure sits on atoms $y_1, \dots, y_m$ with masses $a_1, \dots, a_m$, then a draw is

$$F = \sum_{j=1}^m w_j \delta_{y_j}, \quad (w_1, \dots, w_m) \sim \text{Dirichlet}(a_1, \dots, a_m).$$

- This is nothing but the DP's defining finite-partition property (take the partition that isolates each atom).
- Apply it at a centering draw $\phi \sim \pi(\phi \mid x_{1:n})$, whose DP posterior has (unnormalized) base measure $\sum_{i=1}^n \delta_{x_i} + c f_\phi$ (total mass $c + n$):
- The n **data points** x_i are atoms, each of mass 1;
- The **model** f_ϕ carries continuous mass c
- However we can discretize it with T **pseudo-samples** $x_{n+t} \sim f_\phi$, each of mass c/T (exact as $T \rightarrow \infty$).
- The base then sits on $n + T$ atoms, so the weights are $\text{Dirichlet}(1, \dots, 1, c/T, \dots, c/T)$
- The functional becomes a finite weighted sum $\int \log f_\phi dF = \sum_j w_j \log f_\phi(x_j)$:

Posterior bootstrap – one posterior sample θ :

```
draw center       $\gamma \sim \pi(\gamma \mid x_{\{1:n\}})$            # the parametric Bayes posterior
draw pseudo-data  $x_{\{n+1\}}, \dots, x_{\{n+T\}} \sim f_\gamma$       # T synthetic points; omit when
draw weights      $(w_1, \dots, w_n, w_{\{n+1\}}, \dots, w_{\{n+T\}}) \sim \text{Dirichlet}(1, \dots, 1, c/T, \dots, c/T)$ 
optimize         $\theta = \operatorname{argmax}_\theta \sum_j w_j \log f_\theta(x_j)$ 
return  $\theta$ 
```

repeat B times – fully in parallel

- Each sample is a **randomly-weighted fit** — the same optimization you already know how to do, with random weights.
- The draws are **independent**: embarrassingly parallel, no Markov chain, no burn-in, no mixing diagnostics.

From standard Bayes to the bootstrap

- Look again at the only free, the concentration c in the weights $\text{Dirichlet}(1, \dots, 1, c/T, \dots, c/T)$.
- It interpolates between **two method you may already know**.
- $c \rightarrow \infty$ is just **standard Bayes**. The center $\phi \sim \pi(\phi \mid x_{1:n})$ swamps the data ($F \rightarrow f_\phi$), so $\phi(F) = \phi$ and the output is the ordinary parametric Bayes posterior.
- $c \rightarrow 0$ is then **the (Bayesian) bootstrap**. Since we have pseudo-samples, the weights are $\text{Dirichlet}(1, \dots, 1)$ on the data alone:

$$\phi = \operatorname{argmax}_\phi \sum_{i=1}^n w_i \log f_\phi(x_i), \quad (w_1, \dots, w_n) \sim \text{Dirichlet}(1, \dots, 1).$$

- This is a **Dirichlet-reweighted MLE** which is exactly **Rubin's Bayesian bootstrap**.
- For **moderate** c this interpolates between Bayesian and frequentist uncertainty.
- $c \rightarrow 0$ (**bootstrap**) $\rightarrow$ **frequentist uncertainty**. The posterior spread reproduces the **sampling distribution of the estimator** $\hat{\phi}$ over repeated datasets. Its credible intervals are therefore (asymptotically) **valid frequentist confidence intervals — even when the model is wrong**.
- $c \rightarrow \infty$ (**standard Bayes**) $\rightarrow$ **model-based uncertainty**. The spread is the ordinary parametric posterior, calibrated to the **model** rather than the population — and typically **too narrow under misspecification**.
- **moderate** c — between the two: the model regularizes the bootstrap, trading a little robustness for stability when n is small.

The moral of NPL: uncertainty about ϕ should come from uncertainty about the **population** F_0 , not from the shape of a likelihood (which we know is false). The bootstrap end gives robust frequentist coverage; the Bayes end gives model-based inference; c lets you place yourself between them.

The martingale posterior

The NPL approach is a **special case of a more general one: martingale posterior** (Fong, Holmes & Walker 2023). Let's generalize exactly what we just did.

Generalizing NPL

- What we did: build a posterior over F (an MDP), then push it through $\phi(F)$. Hidden inside is a **one-step predictive**:

$$p(x_{i+1} \mid x_{1:i}, \phi) = \frac{c}{c+i} f_\phi + \frac{1}{c+i} \sum_{j=1}^i \delta_{x_j},$$

- Marginalizing ϕ over its posterior $\pi(\phi \mid x_{1:i})$ then gives the full posterior predictive itself,

$$p(x_{i+1} \mid x_{1:i}) = \underbrace{\frac{c}{c+i} \int f_\phi \pi(\phi \mid x_{1:i}) d\phi}_{\text{parametric posterior predictive}} + \frac{1}{c+i} \sum_{j=1}^i \delta_{x_j}.$$

- Drawing F from the posterior is the same as **iterating this predictive forever**: sample $x_{n+1}, x_{n+2}, \dots$ one at a time, each from the current predictive, and take $F_\infty = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \delta_{x_i}$.
- Now generalize**: Replace this with *any* one-step predictive $p(x_{i+1} \mid x_{1:i})$ and run the same two steps:

$$(1) \text{ impute } X_{n+1:\infty} \sim \prod_{i>n} p(x_i \mid x_{1:i-1})$$

$$(2) \text{ compute } \phi_\infty = \phi(F_\infty).$$

- The randomness in the imputed future *is* the posterior over ϕ .
- This is the **martingale posterior** — the predictive, not a prior $\times$ likelihood, is the modelling object.

Why this is a posterior

- **Uncertainty is the unseen population.** If we saw all of $x_{1:\infty}$, the parameter $\phi_\infty = \phi(F_\infty)$ would be known *exactly*
- So posterior uncertainty about ϕ is exactly uncertainty about the unobserved remainder $x_{n+1:\infty}$.
- **Doob (1949):** if the predictive is the Bayesian posterior predictive, then $\phi_\infty \sim \pi(\phi \mid x_{1:n})$
- Predictive resampling reproduces the ordinary Bayes posterior *exactly*.
- So this **generalizes** Bayes rather than replacing it.
- **The predictive viewpoint.** We specify the predictive $p(x_{i+1} \mid x_{1:i})$ directly, instead of deriving it from a likelihood and prior.
- Predictives can be **checked against future data**, unlike a prior.
- The sequence must satisfy a **martingale condition**: writing $P_i(A) = \Pr(x_{i+1} \in A \mid x_{1:i})$ for the one-step predictive probability of a set A ,

$$\mathbb{E}[P_{i+1}(A) \mid x_{1:i}] = P_i(A) \quad \text{for all } A, i.$$

- Each $P_i(A)$ is then a **martingale**, so by Doob's convergence theorem it has a limit — which is exactly what makes F_∞ exist (equivalently, the sequence is *conditionally identically distributed*; Berti et al. 2020; Fong et al. 2023).

Predictive resampling

In practice we run to a large N rather than ∞ :

Predictive resampling – one martingale-posterior sample:

```
for i = n+1, ..., N      (N large):  
    sample  x_i ~ p(x_i | x_{1:i-1})  
    update  the predictive to include x_i  
return  theta_N = theta(F_N)    # F_N = empirical measure of x_{1:N}  
repeat B times – independent, parallel
```

Conjugate-normal check:

- $f_\phi = \mathcal{N}(\phi, 1)$ and $\pi(\phi) = \mathcal{N}(0, 1)$
- Predictive is $\mathcal{N}(\bar{\phi}_n, 1 + \frac{1}{n+1})$ with $\bar{\phi}_n = \frac{\sum_i x_i}{n+1}$.
- Running the loop, the running mean $\bar{\phi}_N$ drifts to a random limit whose spread matches $\pi(\phi \mid x_{1:n})$.

"Post-Bayes"

NPL / martingale posterior are part of a recent thread that some call **post-Bayesian** or **predictive Bayesian** methods.

- **Why now:** Modern ML is great at building predictive distributions $p(x_{n+1} \mid x_{1:n})$.
- Predictive / post-Bayesian methods seek to center those distributions in developing ways of quantifying uncertainty about population-level parameters.
- Population level parameters $\phi(F)$ are functionals of the population distribution; uncertainty in $\phi(F)$ is driven by uncertainty in F .
- A lot this work resonates/inherits from the **prequential principle (Dawid 1984)**: judge a method by its sequential one-step forecasts $p(x_i \mid x_{1:i-1})$ against data that actually arrive. Predictions are **verifiable** while statements about ϕ are not.

The throughline: a model is a **tool** — a center, a predictive, a loss — and good inference stays honest about the gap between that tool and the world.

References

- Rubin (1981) "The Bayesian bootstrap" (*Ann. Statist.*)
- Lyddon, Walker & Holmes (2018) "Nonparametric learning from Bayesian models with randomized objective functions" (*NeurIPS*)
- Fong, Holmes & Walker (2023) "Martingale posterior distributions" (*JRSS-B*)
- Dawid (1984) "Statistical theory: the prequential approach" (*JRSS-A*)